```
//code eater-for just demonstrate
const ethers = require('ethers');
const abi = require('./contract.json')
require('dotenv').config();

const apiKey =process.env.API_KEY
const privateKey = process.env.PRIVATE_KEY

const readSmartContract = async()=>{
  const provider = new
ethers.JsonRpcProvider(`https://eth-sepolia.g.alchemy.com/v2/${apiKey}`)
  const contractAddress="0x83E8d605e73c0A8Da2F2B3F4046599Fa41514f52"
  const contract = new ethers.Contract(contractAddress,abi,provider);
  const data = await contract.retrieve();
  console.log(Number(data))
}

 readSmartContract()
const writeSmartContract = async()=>{
    const provider = new
ethers.JsonRpcProvider(`https://eth-sepolia.g.alchemy.com/v2/${apiKey}`)
    const contractAddress="0x83E8d605e73c0A8Da2F2B3F4046599Fa41514f52"
    const wallet = new ethers.Wallet(privateKey,provider)
    const contract = new ethers.Contract(contractAddress,abi,wallet);
    const tx = await contract.store(1);
    await tx.wait()
    console.log("successful")
  }
writeSmartContract()
```

```
//from youtube video

"use client"; // This MUST be the first line

import { useEffect, useState } from "react";
import { ethers } from "ethers";
import { abi, contractAddress } from "./constant";
```

```
export default function Home() {
  const [account, setAccount] = useState(null);
  const [provider, setProvider] = useState(null);
  const [name, setName] = useState("");
  const [age, setAge] = useState("");
  const [events, setEvents] = useState([]);

  useEffect(() => {
    const fetchPastEventsAndListen = async () => {
      if (!account || !provider) return;

      const signer = await provider.getSigner();
      const contract = new ethers.Contract(contractAddress, abi, signer);

      // 1⃣ Fetch past UserRegistered events
      try {
        const filter =
//agar tu UserRegistered() yaha pe kuch bhi pass nahi karega to woh sab dega(jab jab
//yeh call hua tha.) aur jab tu address dalega to woh sirf uss address ka dega jab
//jab woh call hua tha.
contract.filters.UserRegistered("0x78c36eD4B3cB79Cb78b187fDAF178DF0C36e5016"); // no
args = all users
        const events = await contract.queryFilter(filter, 0, "latest");

        console.log("📦 Past Events:");
        events.forEach((event) => {
          const { user, name, age } = event.args;
          console.log("🕒", user, name, age.toString());
          setEvents((prev) => [
            ...prev,
            `User: ${user}, Name: ${name}, Age: ${age.toString()}`
          ]);
        });
      } catch (err) {
        console.error("❌ Error fetching past events:", err);
      }

      // 2⃣ Live listener
      contract.removeAllListeners("UserRegistered");
```

```
 //this is for all user(like user is active on its dapp and other user from its dapp
//show that also like(joined now) like that.live means basically like koi aur user
//bhi register kar raha hai to mujhe pata chalega)
      contract.on("UserRegistered", (user, name, age) => {
        console.log("📢 New UserRegistered event:", user, name, age.toString());
        setEvents((prev) => [
          ...prev,
          `User: ${user}, Name: ${name}, Age: ${age.toString()}`
        ]);
      });

      console.log("🎧 Event listener added for UserRegistered");
    };

    fetchPastEventsAndListen();
  }, [account, provider]);
  // re-run only when account/provider changes



  const connectWallet = async () => {
    if (typeof window !== "undefined" && typeof window.ethereum !== "undefined") {
      const provider = new ethers.BrowserProvider(window.ethereum);
      // const provider2 = new
ethers.JsonRpcProvider("https://ethereum-sepolia-rpc.publicnode.com");
      // const provider3 = new
ethers.JsonRpcApiProvider("https://sepolia.infura.io/v3/2de477c3b1b74816ae5475da6d289
208")
      const signer = await provider.getSigner();
      const contract = new ethers.Contract(contractAddress, abi, signer);
      try {
        const accounts = await window.ethereum.request({
          method: "eth_requestAccounts",
        });
        const chainId = await window.ethereum.request({
          method: "eth_chainId",
        });
```

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      // get the chainid in number format
      const chainIdNumber = parseInt(chainId, 16);
      console.log("Connected to chain ID:", chainIdNumber);
      console.log("Connected to account:", accounts[0]);
      console.log("contract:", contract);
      setProvider(provider);
      setAccount(accounts[0]);
    } catch (err) {
      console.error("User rejected connection", err);
    }
  } else {
    alert("Please install MetaMask!");
  }
};


const depositEther = async () => {
  try {
    const provider = new ethers.BrowserProvider(window.ethereum);
    const signer = await provider.getSigner();
    const contract = new ethers.Contract(contractAddress, abi, signer);

    const amountInEther = "0.01";
    const amountInWei = ethers.parseEther(amountInEther);

    const tx = await contract.deposit({
      value: amountInWei, // 👈 This sends ETH
    });

    console.log("⛏️ Mining transaction...");
    const receipt = await tx.wait();
    console.log("✅ Transaction mined:", receipt.transactionHash);

    // Optional: check for event in logs
    const event = receipt.logs
      .map((log) => {
        try {
          return contract.interface.parseLog(log);
        } catch {
          return null;
```

```
          }
        })
        .find((parsed) => parsed && parsed.name === "EtherDeposited");

      if (event) {
        const { from, amount } = event.args;
        console.log(`📥 Deposit event: ${from} sent ${ethers.formatEther(amount)}
ETH`);
      } else {
        console.log("❌ No EtherDeposited event found.");
      }
    } catch (err) {
      console.error("❌ Error during deposit:", err);
    }
  };

  // register user

  const registerUser = async () => {

    const provider = new ethers.BrowserProvider(window.ethereum);
    const signer = await provider.getSigner();

    const contract = new ethers.Contract(contractAddress, abi, signer);
    try {
      const tx = await contract.registerUser(name, parseInt(age));
      await tx.wait();
      console.log("Transaction hash:", tx.hash);
      console.log("User registered:", name, age);
      //yeh jab registereduser call hoga tab yeh event chalega(one time-when this is
//called.and also we can fetched required event threw filter.)
      const event = tx.logs
        .map((log) => {
          try {
            return contract.interface.parseLog(log);
          } catch {
            return null;
          }
        })
        .filter((parsed) => parsed && parsed.name === "UserRegistered")[0];
```

```
    if (event) {
      const { user, name, age } = event.args;
      console.log("🟢 Event Emitted:");
      console.log("User:", user);
      console.log("Name:", name);
      console.log("Age:", age.toString());
    } else {
      console.log("❌ No UserRegistered event found in logs.");
    }
  }
  catch (err) {
    console.error("Error registering user:", err);
  }


}

return (
  <main style={{ padding: "2rem" }}>
    <h1>Hello in frontend</h1>
    {account ? (
      <p>Connected: {account}</p>
    ) : (
      <button onClick={connectWallet}>Connect MetaMask</button>
    )}
    <div style={{ marginTop: "1rem" }}>
      <input
        type="text"
        placeholder="Enter name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <input
        type="number"
        placeholder="Enter age"
        value={age}
        onChange={(e) => setAge(e.target.value)}
      />
      <button onClick={registerUser}>Register</button>
```

```
        <button onClick={depositEther}>depoiteETH</button>
      </div>
    </main>
  );
}
```